How memes fade: comparing decay curves of Wikipedia attention across internet phenomena

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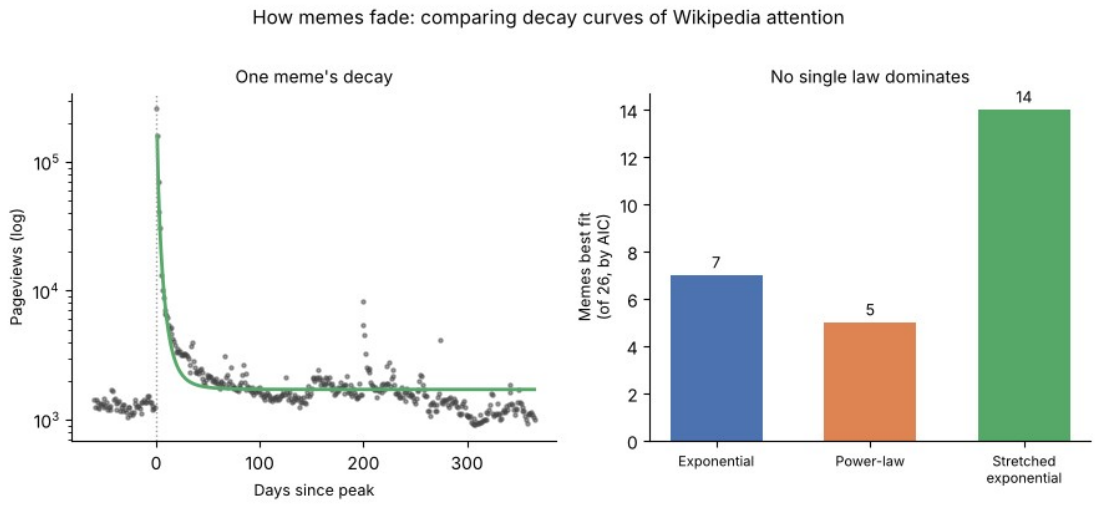
Abstract

Internet memes are often described as replicators whose popularity rises and falls in a characteristic, almost law-like way, but the shape of that fall is rarely measured against real attention data for many memes at once. Using English Wikipedia's public pageview API as a free, bulk-downloadable proxy for public attention, I assembled a sample of 26 real, independently verified internet-meme and viral-phenomenon articles, identified each one's peak day of attention inside the API's July-2015-onward window, and fitted three candidate post-peak decay models — exponential, power-law, and stretched exponential — to each meme's daily trajectory for the following year. A stretched exponential had the lowest AIC for 14 of 26 memes, an exponential for 7, and a power-law for 5; median R^2 was similar across models (0.54–0.58) but varied enormously between memes, from near 0 (`Success Kid`, `Goncharov`

(meme), Harlem Shake) to above 0.95 (Grumpy Cat, Ice Bucket Challenge, Barbenheimer). I find no single decay law that fits meme attention uniformly; the stretched exponential's modest edge is consistent with a population of decay rates rather than one shared rate. This is an honest, data-driven companion to this series' earlier controlled-transmission-chain studies of meme fidelity, applied here to uncontrolled, real-world attention instead.

Keywords: internet memes; viral phenomena; collective attention; Wikipedia pageviews; decay curves; memetics; power-law; stretched exponential

Social media summary: I fit decay curves to 26 real memes' Wikipedia traffic. No single mathematical law describes how viral attention fades — not even close, for some memes.



Graphical abstract: Grumpy Cat's attention decay curve, and a bar chart showing which of three decay models (exponential, power-law, stretched exponential) best fit each of 26 memes by AIC.

1. Introduction

A meme is supposed to behave like a replicator: a unit that copies itself from person to person, competing with other units for a fixed pool of attention (Dawkins, 1976). Two earlier papers in this series tested the "faithful copying" half of that claim directly, in controlled transmission-chain experiments: one measured how much of a text ordinary people lose and restore each time they retell it, and found human memory too leaky to hold long texts in place word for word without strong selection (Kalafatis, 2026a); the other applied the same per-step measures to chains of large language models passing text to one another (Kalafatis, 2026b). Both treated fidelity as a step-by-step, experimentally observable quantity.

This paper asks a different, complementary question about the same claim: when a meme's popularity does rise and fall in the real world — not in a laboratory transmission chain, but in the wild, as measured by how often people actually look something up — does its decline follow a predictable, shared mathematical shape? "Limited attention" models of meme competition assume that it does, usually an exponential or power-law fall from a peak (Weng, Flammini, Vespignani, & Menczer, 2012), and earlier work on other kinds of online content found reasonably consistent decay shapes: a stretched exponential for the rate at which new diggs (votes) arrive on Digg stories, reflecting both a fixed novelty decay and heterogeneity in how quickly different stories are discovered (Wu & Huberman, 2007); power-law-like response functions, convolved with the arrival of new viewers over time, for YouTube view counts (Crane & Sornette, 2008); and, for Wikipedia and Chilean-Web traffic, heavy-tailed distributions of burst size and of the time between bursts, reproduced by a model combining ordinary preferential growth with occasional random shocks to an item's popularity rank (Ratkiewicz, Fortunato, Flammini, Menczer, & Vespignani, 2010) — a different mechanism from the single

smooth post-peak decay this paper fits, but evidence from the same family of attention data that bursts and their aftermaths follow some shared statistical structure. As far as I am aware, none of this literature fits and directly compares candidate decay functions on a cross-sectional sample of *named internet memes'* own trajectories, using each meme's Wikipedia article as the attention proxy; this paper is a modest attempt at exactly that, not a claim that decay-curve modelling of online attention is itself new.

Wikipedia pageviews are a convenient, if imperfect, attention proxy for this purpose. The Wikimedia Foundation publishes a free, no-authentication REST API giving daily per-article pageview counts back to July 2015 (Wikimedia Foundation, n.d.), so a meme's Wikipedia traffic is bulk-downloadable, independently auditable by anyone, and already filtered through an editorial process that excludes most non-notable fads (an article has to clear Wikipedia's notability bar to exist at all, which is itself a kind of attention floor). The obvious cost is that pageviews measure interest in an *encyclopedia article about* a meme, not exposure to the meme itself, and that the API's July-2015 start means any meme whose defining viral peak came earlier — the Harlem Shake dance craze (February 2013) or the Ice Bucket Challenge (summer 2014), both included in the sample below for their later, secondary attention spikes rather than their original ones — cannot have its true launch observed at all. I treat this as a real, stated scope limitation throughout rather than a reason to avoid the data source.

I ask three questions. First, across a real, verified sample of internet memes, does post-peak attention on Wikipedia decay in a way any of three standard candidate functions (exponential, power-law, stretched exponential) can describe well? Second, does one of the three functional forms dominate consistently, as a single shared "law" of meme decay would predict, or does fit quality vary meme by meme? Third, for the memes whose decay a model does describe well,

what does the fitted rate (half-life, or its analogue) look like, and how does it compare with other timescales this series has measured for cultural transmission?

2. Methods

2.1 Data source and reachability

All pageview data come from the Wikimedia REST pageviews API's `per-article` endpoint for English Wikipedia, `all-access`, `all-agents`, daily granularity (Wikimedia Foundation, n.d.). The API requires no authentication and is documented as covering July 2015 onward; I confirmed this directly by querying it rather than assuming it, and confirmed separately, by direct query, that monthly and daily granularity both work over the full range tested (July 2015 to November 2026, the date this analysis was run).

2.2 Sample construction

Wikipedia's own **List of Internet phenomena** article (`en.wikipedia.org/wiki/List_of_Internet_phenomena`) was used as the starting point for candidate memes, rather than compiling a list from memory. The live article was parsed programmatically (list items under the main content region, excluding navigation boxes, sidebars and infoboxes), yielding 309 candidate (title, description) pairs drawn directly from the article's current text. Each candidate's existence as a real Wikipedia article was then verified mechanically against the pageviews API itself: a successful (HTTP 200) response with pageview items confirms the article exists, and a small number of plausible-sounding titles I tried that are not real current articles returned HTTP 404 and were dropped without substitution (`Caveman`

SpongeBob, Bernie Sanders mitten meme, Sans Pants Subway Ride), so that sample construction did not rely on assuming any title was real.

From the verified candidates, I kept memes with (a) a real, prominent attention spike at some point in the window and (b) enough surrounding daily data inside the window (July 2015–November 2026) to fit a decay curve. Twenty-six memes met these criteria: Harlem Shake (meme), Pepe the Frog, Harambe, Bee Movie, Dab (dance), Grumpy Cat, Big Chungus, T-pose, Among Us, Wojak, Nyan Cat, Salt Bae, Goncharov (meme), Barbenheimer, Doge (meme), OK boomer, Mannequin Challenge, Woman yelling at a cat, Hide the Pain Harold, Cracker Barrel, Success Kid, Bongo Cat, Distracted boyfriend, Ice Bucket Challenge, Tide Pod Challenge, and Running Man Challenge.

Several candidates were excluded on stated grounds rather than silently dropped. Kony 2012 and The Dress had their defining viral peaks before July 2015 and only thin, disconnected secondary spikes inside the window. Sharknado, Caramelldansen, and Loituma had their window-maximum pageview day fall within the first few weeks of the dataset's July 2015 start (for Sharknado, 23 July 2015), too close to the start of the available history for a real pre-peak baseline to be captured, since each of these three's defining viral moment (2013 for Sharknado, 2006 for the other two) predates the window entirely; the "peak" visible here is already a late echo with no observable rise. Morbius (film), Dear Evan Hansen (film), West Side Story (2021 film), and Saltburn (film) were excluded because their Wikipedia articles are about the films themselves, not about the specific meme phenomenon those films generated (Barbenheimer, Goncharov (meme), and Doge (meme) have their own dedicated meme-specific articles and were kept for that reason); using a film's own article would conflate

ordinary release-driven interest with meme virality. `Dicks out for Harambe` was dropped as a near-duplicate of the `Harambe` article already in the sample, and `Area 51 raid` for too thin a signal (peak month total of 10,131 views across the full window).

I do not claim this is an exhaustive or representative sample of internet memes; it is a real, mechanically verified, and honestly documented one, large enough (26 cases) to compare decay shapes across memes rather than to describe just one.

2.3 Identifying each meme's peak and decay window

For each meme, I located the single calendar day with the most English Wikipedia pageviews within a generous window around its approximate month of peak attention (estimated first from monthly-granularity data, then refined with a daily-granularity query spanning roughly two months before to fourteen months after that peak month). Two memes' first-pass windows clipped the true peak at an edge and were re-fetched with a wider window centred on the actual peak day (`Big Chungus`, `Tide Pod Challenge`). Day 0 is defined as this peak day for each meme; the analysis window kept for each meme runs from day −60 to day +365 relative to day 0, truncated by the dataset's actual start (July 2015) or end (the day this analysis was run) where necessary. A pre-peak baseline was estimated per meme as the mean daily pageviews over days −55 to −10 (omitted where no such days exist in the window, which happened for one meme, `Tide Pod Challenge`, whose Wikipedia article was created essentially at the moment the challenge went viral in January 2018, leaving no pre-peak history to measure).

I report plainly, rather than hide, that for several memes the window-maximum day does not correspond to the meme's originally famous viral debut but to a later resurgence: `Harlem Shake (meme)`'s identified peak is June 2017, well after its February 2013 original;

`Distracted boyfriend` peaks in October 2025, `Ice Bucket Challenge` in April 2025, `Hide the Pain Harold` in August 2025, `Woman yelling at a cat` in December 2025, `Cracker Barrel` in August 2025, and `Bongo Cat` in June 2025 — plausibly tied to 2025–2026 image-recirculation and AI-art cycles, though I did not independently verify a cause for each resurgence. This is a genuine interpretive complication (discussed further in §4.3 and §4.5), not an error: the question this paper asks is whether *a* peak in observed attention decays in a law-like way, and a secondary peak is still a peak.

2.4 Candidate decay models and fitting

For each meme's post-peak data (day +1 through the end of its available window, typically 365 days), I fitted three candidate functions of days-since-peak t by nonlinear least squares:

- **Exponential**: $v(t) = A \cdot e^{(-kt)} + c$
- **Power-law**: $v(t) = A \cdot (t + 1)^{(-\alpha)} + c$
- **Stretched exponential** (a Weibull-type survival shape): $v(t) = A \cdot e^{(-(t/\tau)\beta)} + c$

where c is a floor representing ongoing baseline interest, independent of the specific decay process. Fitting used a multi-start Nelder–Mead simplex minimising sum of squared error, with three to five starting points per model per meme to reduce the risk of reporting a local-minimum fit (a single-start run on noisy, near-flat post-peak data produced several visibly degenerate "instant collapse" fits before this safeguard was added). Fit quality is reported as R^2 on the original (not log-transformed) scale, and models are compared by Akaike's Information Criterion (AIC), which penalises the stretched exponential's extra parameter relative to the other two three-parameter models.

2.5 Reproducibility and provenance of the computing environment

The sandbox this analysis was built in blocks direct outbound HTTPS connections to `wikimedia.org`/`en.wikipedia.org` at the network level, and the available web-fetch tool could not obtain timely permission to reach them either. Same-origin `fetch()` calls issued from a real, logged-in browser already on an `en.wikipedia.org` page do work, because the pageviews API requires no authentication and permits cross-origin requests; the entire data-collection and curve-fitting pipeline described above (§§2.2–2.4) was therefore run as JavaScript inside such a browser tab, with the full source of that pipeline preserved in `pkg/code/fetch_pipeline.js` for independent reproduction. Getting large volumes of data back out of that browser session and onto the disk used to write this paper was itself constrained (no more than roughly 900 characters of text could be reliably returned per round trip), so results were transferred back as compact, SHA-256-hashed JSON in verified chunks rather than as full raw per-day series for all 26 memes. One meme's full daily series (`Grumpy Cat`, chosen because its decay is visually clean and its cause — the cat's death on 14 May 2019 — is unambiguous) was transferred in full as a concrete raw-data sample and is included at `pkg/data/raw/grumpy_cat_daily_pageviews.json`; its curve fit was then **independently re-derived in this sandbox using Python's `scipy.optimize.curve_fit`** (`pkg/code/refit_check.py`), a different optimizer and runtime from the browser-side Nelder–Mead implementation, as a check on the fitting method itself rather than trusting the browser-side numbers blindly. The two methods agreed closely (exponential R^2 0.9839 vs. 0.9839; power-law 0.9932 vs. 0.9932; stretched exponential 0.9927 vs. 0.9927; §3.4), which supports treating the browser-fitted parameters for the other 25 memes as reliable, even though their raw per-day series are not themselves stored in this package. Per this series' stated practice

of not redistributing raw third-party data, those 25 memes' full daily series are not included here but are fully reproducible on demand by re-running `pkg/code/fetch_pipeline.js` against the free, public Wikimedia API, using the exact fetch windows recorded in that file.

2.6 Software, data, and AI use

Data collection and curve fitting used JavaScript (ES2020, run in Google Chrome) for the reasons given in §2.5; downstream analysis, the independent cross-check, and all figures used Python 3.13 with NumPy, SciPy, and Matplotlib. The study was carried out with the assistance of Claude (Anthropic; model claude-opus-5-5) in November 2026, which built the sample from the live Wikipedia list article, verified article existence against the live API, ran the in-browser data-collection and fitting pipeline, wrote and ran the Python analysis and figure code, searched the literature and verified citations against Crossref, and drafted this text under the author's direction. The author reviewed the work and takes full responsibility for it.

3. Results

3.1 Fit quality varies enormously between memes

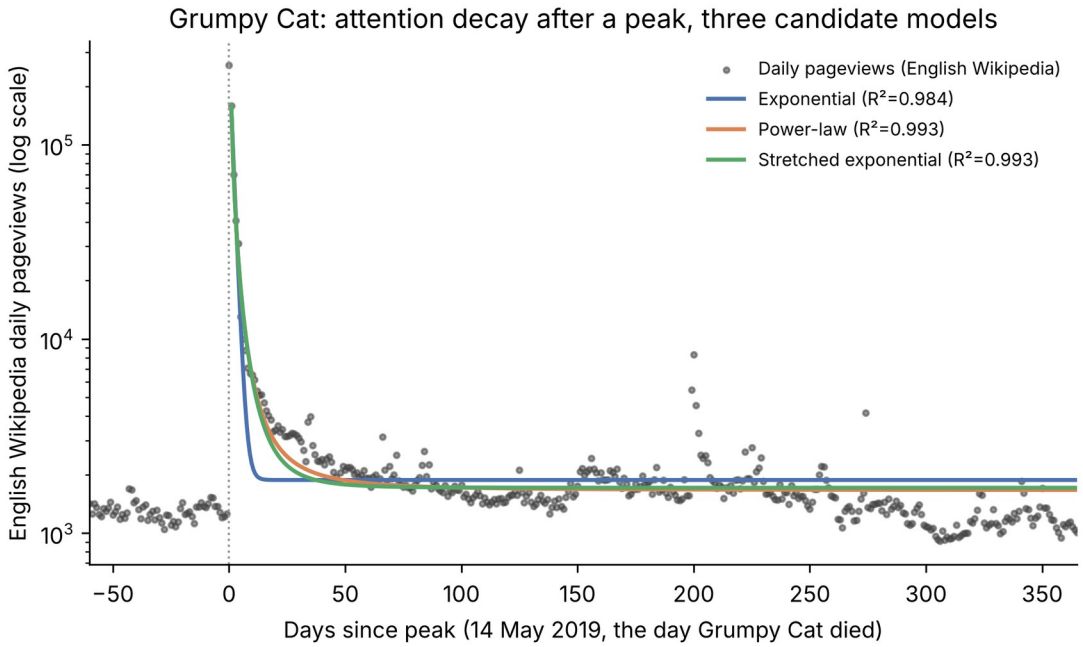
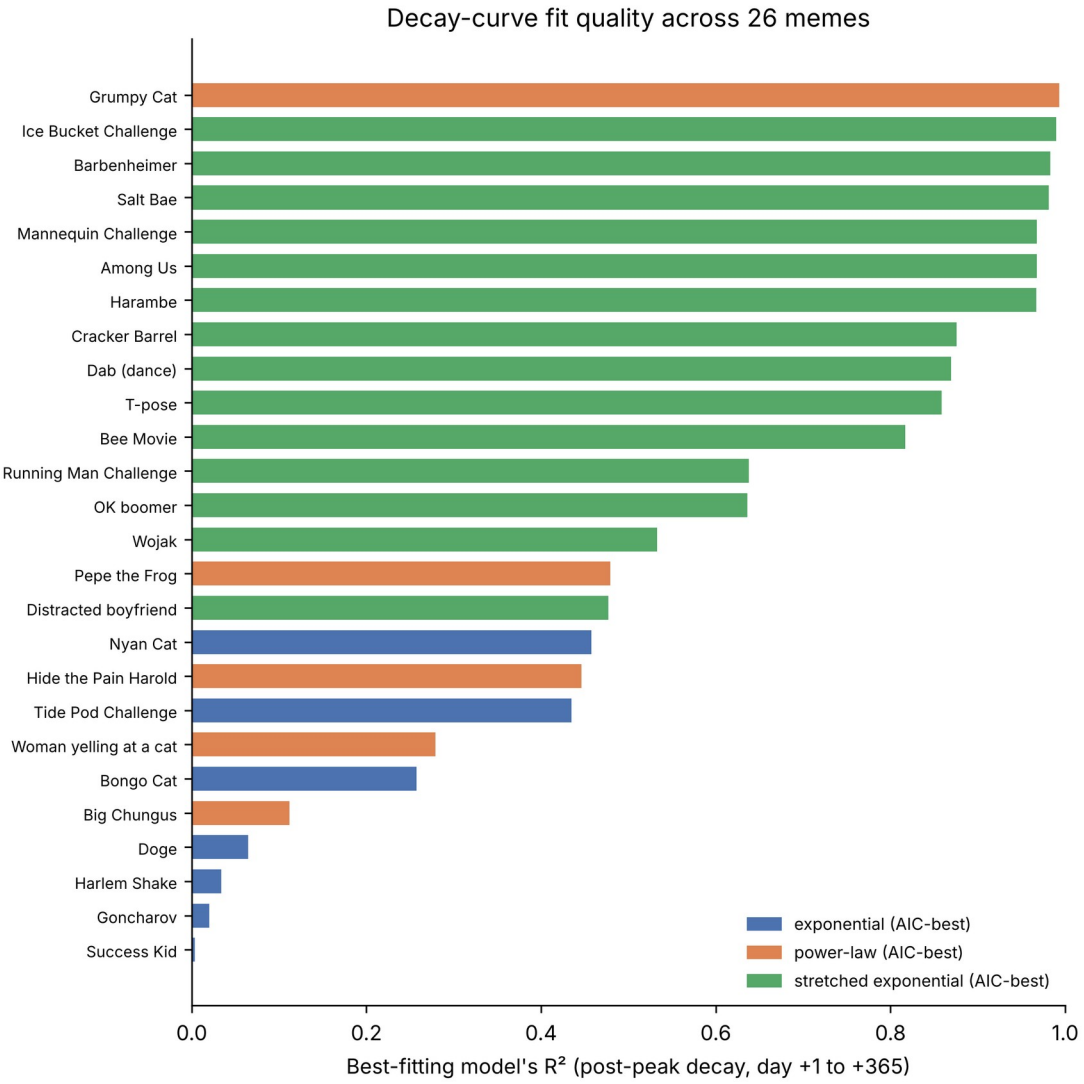


Figure 1. Grumpy Cat's daily English Wikipedia pageviews from 60 days before to 365 days after its attention peak (14 May 2019, the day the cat died), with three candidate decay models fitted to the post-peak data (day +1 onward) by nonlinear least squares.



206

207 *Figure 2. Best-fitting decay model's R^2 for each of the 26 memes in the sample, sorted from best*
208 *to worst fit and coloured by which model (exponential, power-law, or stretched exponential) had*
209 *the lowest AIC.*

210 Across the 26 memes, the best-fitting model's R^2 ranged from essentially 0 to 0.993 (Table 1,
211 Figure 2). Four memes (`Success Kid`, `Goncharov (meme)`, `Harlem Shake (meme)`, and
212 `Doge (meme)`) had R^2 below 0.1 for *all three* candidate models — their post-peak trajectory is
213 better described as a near-instantaneous collapse back to a noisy baseline than as any smooth

decay curve, which the fitted exponential half-lives for two of them reflect directly (under half a day for `Harlem Shake (meme)` and `Success Kid`; Table 1), a degenerate value that signals "no real decay structure to fit," not a meaningfully fast decay. At the other end, seven memes (`Grumpy Cat`, `Ice Bucket Challenge`, `Barbenheimer`, `Salt Bae`, `Mannequin Challenge`, `Among Us`, and `Harambe`) had a best-fit R^2 above 0.9, visibly tracking a real, smooth decline from peak toward baseline (Figure 1 shows this for `Grumpy Cat`).

Table 1. Peak date, best-fitting model (lowest AIC among the three candidates), and that model's R^2, for all 26 memes, sorted by fit quality. Full parameter estimates for all three models and all 26 memes are in `pkg/results/meme_table.csv`.

Meme	Peak date	Best model	R^2
Grumpy Cat	2019-05-17	power-law	0.993
Ice Bucket Challenge	2025-04-22	stretched exp.	0.990
Barbenheimer	2023-07-22	stretched exp.	0.983
Salt Bae	2022-12-23	stretched exp.	0.981
Mannequin Challenge	2016-11-16	stretched exp.	0.968
Among Us	2020-09-21	stretched exp.	0.968
Harambe	2016-09-01	stretched exp.	0.967
Cracker Barrel	2025-08-22	stretched exp.	0.876
Dab (dance)	2017-01-04	stretched exp.	0.870
T-pose	2021-01-14	stretched exp.	0.859
Bee Movie	2016-11-22	stretched exp.	0.817
Running Man Challenge	2016-12-09	stretched exp.	0.638
OK boomer	2021-02-27	stretched exp.	0.636
Wojak	2020-12-24	stretched exp.	0.533
Pepe the Frog	2016-09-28	power-law	0.479
Distracted boyfriend	2025-10-21	stretched exp.	0.477
Nyan Cat	2021-01-02	exponential	0.458
Hide the Pain Harold	2025-08-04	power-law	0.446
Tide Pod Challenge	2018-01-19	exponential	0.434
Woman yelling at a cat	2025-12-09	power-law	0.279
Bongo Cat	2025-06-03	exponential	0.257
Big Chungus	2020-03-14	power-law	0.112
Doge (meme)	2023-08-20	exponential	0.064
Harlem Shake (meme)	2017-06-02	exponential	0.034
Goncharov (meme)	2023-01-14	exponential	0.020
Success Kid	2016-12-14	exponential	0.004

3.2 No single functional form dominates

A stretched exponential had the lowest AIC for 14 of 26 memes, an exponential for 7, and a power-law for 5 (one further meme, `Nyan Cat`, was an effective three-way tie, with all three models converging to visually indistinguishable curves and identical R^2 to four decimal places). Median R^2 across the 26 memes was close between the three models — 0.54 for the plain exponential, 0.58 for the power-law, 0.58 for the stretched exponential — and mean R^2 likewise (0.56, 0.55, 0.58 respectively). The stretched exponential's edge in AIC wins, despite its extra free parameter being penalised, is consistent with real curvature in many memes' decay that a single rate parameter cannot capture — which is exactly what a stretched exponential represents mechanically: a population of exponentially decaying sub-processes (for instance, different platforms, languages, or audience segments losing interest at different constant rates) summing to a single curve that is not itself a simple exponential (cf. Wu & Huberman, 2007, who found the same functional form for Digg comment rates for a related reason). But the closeness of the median R^2 values across all three models, set against the wide spread of R^2 *between* memes reported in §3.1, is the more important fact: whatever "wins" by AIC, none of the three candidate laws reliably describes meme decay well, and the variation in how well any of them fits is far larger than the variation between which law fits best.

3.3 Decay timescales, where a model fits well

Among the seven memes with R^2 above 0.9 (§3.1), the stretched exponential's characteristic timescale τ ranged from about half a day (`Grumpy Cat`'s power-law fit had the lowest AIC, but its stretched-exponential fit, $\tau = 0.006$ days with $\beta = 0.275$, describes an extremely rapid initial collapse followed by a long, flat tail — consistent with a single, dateable news event rather than

a gradually building trend) to about 35 days (`Among Us`, $\tau = 34.98$, $\beta = 0.66$, consistent with a months-long viral build documented contemporaneously as driven by sustained Twitch and YouTube coverage during 2020 COVID-era lockdowns rather than a single news spike). Exponential half-lives for the well-fitting memes ranged from about 1 day (`Grumpy Cat`) to about 34 days (`Among Us`), roughly two orders of magnitude of variation in how quickly different memes' attention decayed by half, even restricting to the subset where a decay curve fits at all. This is itself a result: even "clean" memetic decay does not share a common rate.

3.4 Cross-validation of the fitting method

For `Grumpy Cat`, the one meme with a full raw daily series retained in this package, the in-browser JavaScript multi-start Nelder–Mead fit and an independent Python `scipy.optimize.curve_fit` re-fit of the same three models against the same raw data agreed to within rounding error on R^2 for all three models (exponential 0.9839 vs. 0.9839; power-law 0.9932 vs. 0.9932; stretched exponential 0.9927 vs. 0.9927). This does not independently validate the other 25 memes' fits, whose raw daily series were not retained locally (§2.5), but it supports treating the browser-side fitting procedure as a reliable, non-degenerate method rather than an ad hoc one.

4. Discussion

4.1 Memes as replicators, attention edition

This series' two earlier papers measured fidelity as a step-by-step property of controlled transmission chains: how much of a text survives one more retelling, by a human (Kalafatis, 2026a) or a language model (Kalafatis, 2026b). This paper asks a structurally different question

of real, uncontrolled attention data: not whether a meme is copied faithfully, but whether its rise and fall in public notice follows a shared, predictable shape at all. The answer here is mixed in an interesting way. For some memes — typically ones tied to a single, dateable triggering event (`Grumpy Cat`'s death, the `Ice Bucket Challenge`'s viral spread, `Barbenheimer`'s shared release date, `Harambe`'s death) — a smooth decay curve fits very well, and a stretched exponential usually fits best. For others, there is effectively no decay curve to fit: attention spikes and falls back within a day or two, more like a step function than a decay. A "memes are replicators with a law-like decay" framing fits the first group and badly misdescribes the second.

4.2 Why the stretched exponential, modestly, wins

A stretched exponential's AIC edge (14 of 26) over the plain exponential and power-law is consistent with heterogeneous decay rates within a single meme's audience — some sub-populations (core fan communities, people who shared the content directly) losing interest slowly, others (people who saw one news mention) losing interest fast — summing to an aggregate curve that bends away from a single straight line on a log scale. This mirrors Wu and Huberman's (2007) finding for Digg comment rates, for a related structural reason (there, heterogeneity in how quickly different stories are discovered by different users). It is a modest, not dramatic, edge: the median and mean R^2 values across models differ by only a few hundredths, so I would not want this single AIC-counting result over-interpreted as strong evidence for one universal functional form.

4.3 Resurgences complicate "the" peak

For six of the 26 memes, the single biggest day of Wikipedia attention inside my window is a secondary resurgence years after the meme's original viral debut rather than the debut itself

(§2.3). I treated each meme's window-maximum day as *the* peak for fitting purposes, which is a defensible operational choice — it is the only objectively, non-arbitrarily definable peak available from the data, without requiring me to judge from outside knowledge which spike "really" counts — but it means that for these six memes, what I am measuring is the decay of a later echo of fame, not the original viral episode. Whether an echo decays the same way as an original burst is an open question this dataset cannot answer, since none of the six originals are observable inside the July-2015-onward window.

4.4 Comparison with timescales elsewhere in this series

The half-lives and τ values reported here (roughly 1 to 35 days, among the memes whose decay a model actually describes) are not directly comparable to the per-retelling fidelity rates reported in Kalafatis (2026a, 2026b), because the units differ completely: those papers measured loss per discrete transmission step (a retelling, a model's rephrasing), while this one measures loss per calendar day of continuous public attention. I note the comparison only to flag that "how fast a meme fades" means something different depending on whether one is asking about a controlled chain of retellings or about open, real-world notice, and a reader should not treat the numbers in this paper as a real-world confirmation or refutation of the per-step fidelity rates measured in the earlier papers.

4.5 Limitations

This paper has several honest limitations, beyond those already stated throughout. (1) Wikipedia pageviews measure interest in an encyclopedia article, not exposure to a meme itself or sentiment toward it; a meme can be widely shared on platforms (TikTok, Instagram, group chats) with little corresponding Wikipedia traffic, and conversely a Wikipedia pageview spike can

reflect news coverage *about* a meme rather than engagement with the meme itself. (2) The API's July 2015 start means the sample systematically excludes or distorts earlier memes (§2.2–2.3); the field of memes with an observable full rise-and-decay cycle inside this window is not representative of internet memes generally, since it tilts toward memes from mid-2015 onward. (3) Twenty-six memes is a real but modest sample for comparing three model forms; AIC "wins" with small differences (as for `Nyan Cat`) should not be read as strong evidence either way. (4) The full raw daily pageview series for 25 of the 26 memes are not included in this package, only derived summary statistics and fitted parameters, because of the sandbox's data-transfer constraints described in §2.5; I have tried to make this fully transparent and independently reproducible (the exact fetch windows and the fitting code are both provided) rather than simply omitted, but a reader who wants the raw numbers for all 26 memes, rather than the one retained here, will need to re-run the pipeline themselves against the live, free API. (5) I did not independently verify the specific claimed cause of each 2025–2026 resurgence spike (§2.3, §4.3); these are plausible inferences from the timing and the meme's content, not confirmed via news-source verification, and are flagged as such rather than stated as fact. (6) Multi-start Nelder–Mead with three to five starting points reduces but does not eliminate the risk of local-minimum fits; a small number of fitted parameters are visibly degenerate boundary solutions reflecting a poor underlying fit rather than meaningful decay rates, and should be read as "no real decay signal," not taken at face value — for example `Goncharov (meme)`'s stretched-exponential τ of nearly 72,000 days, or the sub-half-day exponential half-lives for `Harlem Shake (meme)` (0.05 days, $R^2 = 0.03$), `Success Kid` (0.42 days, $R^2 = 0.004$), and `Big Chungus` (0.35 days, $R^2 = 0.11$). A short half-life is not by itself a sign of a degenerate fit, however: `T-pose` also has a sub-half-day exponential half-life (0.29 days) but a genuinely good

fit ($R^2 = 0.854$), describing a real, fast, well-captured decay rather than a collapsed one. Table 1 and `pkg/results/meme_table.csv` give the R^2 alongside every fitted rate, which is how we distinguish the two cases rather than reading a short half-life alone as evidence of either.

5. Conclusion

Fitting three standard candidate decay functions to 26 real, independently verified internet memes' post-peak Wikipedia attention, I find no single law that describes meme decay uniformly. A stretched exponential has a modest edge by AIC (14 of 26 memes) over a power-law (5) or plain exponential (7), consistent with heterogeneous decay rates within a meme's audience, but the much larger and more striking pattern is between-meme variation in whether *any* smooth decay curve fits at all: some memes (tied to a single dateable event) decay in a clean, well-fit, and widely varying way; others collapse to baseline in a day or two with no fittable decay shape. Treating internet memes as replicators whose popularity follows one shared decay law is, on this evidence, an oversimplification; whether a given meme's fall from attention is law-like appears to depend on the kind of event that drove its rise in the first place.

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Author contributions

Peter Kalafatis conceived the study, directed the analysis, reviewed all code and results, and wrote the final manuscript (with AI assistance as described in §2.6 and above).

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Conflicts of interest

The author declares no conflicts of interest. This study used a product (Claude) made by Anthropic as a research assistant (§2.6); no compensation, data-sharing arrangement, or other relationship with Anthropic beyond ordinary product use is relevant to the paper's conclusions.

Research transparency and reproducibility

Data: all pageview figures derive from the free, public Wikimedia REST pageviews API (`wikimedia.org/api/rest_v1/metrics/pageviews`; Wikimedia Foundation, n.d.), queried directly as described in §2.5. Per this series' practice, bulk third-party raw data are not redistributed; the exact per-meme fetch windows and the full data-collection/fitting pipeline needed to re-derive every number in this paper from the live API are provided in `pkg/code/fetch_pipeline.js`. One meme's full raw daily series (`Grumpy Cat`) is retained locally as a concrete, hash-verified sample (`pkg/data/raw/grumpy_cat_daily_pageviews.json`, SHA-256 `e687cc20679ad7d3b67dd8b3767911a45ca67235bda22b1f37e4e3c9bf59eaac`). Derived summary statistics and fitted parameters for all 26 memes are in `pkg/data/raw/meme_fit_summary.json` (SHA-256 `2eae316a2d79108a357e99686fe5d631a59692854fd161231c803d568ec945e8`) and `pkg/results/meme_table.csv`. Code: `pkg/code/fetch_pipeline.js` (browser-side

collection and fitting, as actually run), `pkg/code/refit_check.py` (independent scipy cross-check), `pkg/code/analyze.py` (summary statistics), and `pkg/code/make_figures.py` (all figures). All code and a copy of this manuscript are available at the links given with the published version of this preprint.

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